C++

Lesson 1 : Language Fundamentals

Lesson Coverage

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Lesson Coverage

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- Pointers
- References
- Basic Input/Output
- Data Structures

Overview

- C++ is a statically typed, compiled, general-purpose, case-sensitive, free-form programming language that supports procedural, object-oriented, and generic programming.

- C++ is regarded as a middle-level language, as it comprises a combination of both high-level and low-level language features.

- C++ was developed by Bjarne Stroustrup starting in 1979 at Bell Labs in Murray Hill, New Jersey

- C++ is a superset of C and any legal C program is a legal C++ program.

Overview

Object-Oriented Programming

- C++ fully supports object-oriented programming, including the four pillars of object-oriented development:
  - Object
  - Class
  - Encapsulation
  - Inheritance
  - Polymorphism
  - Abstraction

Overview

Standard Libraries

- Standard C++ consists of three important parts:
  - The core language giving all the building blocks including variables, data types and literals, etc.
  - The C++ Standard Library giving a rich set of functions manipulating files, strings, etc.
  - The Standard Template Library (STL) giving a rich set of methods manipulating data structures, etc.

Overview

Use of C++

- C++ is used by hundreds of thousands of programmers in essentially every application domain.

- C++ is being highly used to write device drivers and other softwares that rely on direct manipulation of hardware under realtime constraints.

- C++ is widely used for teaching and research because it is clean enough for successful teaching of basic concepts.

Environment Setup

- Editor & Compiler -
  - Turbo C++
  - Dev C++
  - Eclipse C++
  - Visual Studio 2008/2010
  - And many more ….

C++ Program Structure

```
#include <iostream>
using namespace std;
//class definition can go here
// main() is where program execution begins.

int main()
{
   cout << "Hello World"; // prints Hello World
   return 0;
}
```

Comments

- Single line

```
//  comments
```

- Multiline

```
/*
  comment1
  comment2

*/
```

Data Types

Type	Typical Bit Width	Typical Range
char	1byte	-127 to 127 or 0 to 255
unsigned char	1byte	0 to 255
signed char	1byte	-127 to 127
int	4bytes	-2147483648 to 2147483647
unsigned int	4bytes	0 to 4294967295
signed int	4bytes	-2147483648 to 2147483647
short int	2bytes	-32768 to 32767
unsigned short int	Range	0 to 65,535
signed short int	Range	-32768 to 32767
long int	4bytes	-2,147,483,647 to 2,147,483,647
signed long int	4bytes	same as long int
unsigned long int	4bytes	0 to 4,294,967,295
float	4bytes	+/- 3.4e +/- 38 (~7 digits)
double	8bytes	+/- 1.7e +/- 308 (~15 digits)
long double	8bytes	+/- 1.7e +/- 308 (~15 digits)
wchar_t	2 or 4 bytes	1 wide character

Data Types

```
#include <iostream>
using namespace std;
int main()
{
   cout << "Size of char : " << sizeof(char) << endl;
   cout << "Size of int : " << sizeof(int) << endl;
   cout << "Size of short int : " << sizeof(short int) << endl;
   cout << "Size of long int : " << sizeof(long int) << endl;
   cout << "Size of float : " << sizeof(float) << endl;
   cout << "Size of double : " << sizeof(double) << endl;
   cout << "Size of wchar_t : " << sizeof(wchar_t) << endl;
   return 0;
}
```

Typedefs

- Creating a new name for an existing type using typedef.
- Following is the simple syntax to define a new type using typedef:
  - typedef type newname;

- For example, the following tells the compiler that feet is another name for int:
  - typedef int feet;

- Now, the following declaration is perfectly legal and creates an integer variable called distance:
  - feet distance;

Variable Declaration

- Variable Declaration
  - int i, j, k;
  - char c, ch;
  - float f, salary;
  - double d;

- Variable Initialization/declaration time
  - int d = 3, f = 5; // declaration of d and f.
  - int d = 3, f = 5; // definition and initializing d and f.
  - byte z = 22; // definition and initializes z.
  - char x = 'x'; // the variable x has the value 'x'

Variable Scope

- A scope is a region of the program and broadly speaking there are three places, where variables can be declared:
  - Inside a function or a block which is called local variables,
  - In the definition of function parameters which is called formal parameters.
  - Outside of all functions which is called global variables.

Variable Scope

```
#include <iostream>
using namespace std;
// Global variable declaration:
int g;
int main ()
{
  // Local variable declaration:
  int a, b;
  // actual initialization
  a = 10;
  b = 20;
  g = a + b;
  cout << g;
  return 0;
}
```

Storage Classes

- A storage class defines the scope (visibility) and life-time of variables and/or functions within a C++ Program.
- These specifiers precede the type that they modify.
- There are following storage classes, which can be used in a C++ Program
  - auto
  - register
  - static
  - extern
  - mutable

- The **auto** Storage Class
- The **auto** storage class is the default storage class for all local variables.

Using auto Storage class

- ```
  #include <iostream>
  using namespace std;

  int main() {
  auto x = 10; // int
  auto y = 3.14; // double
  auto name = "C++"; // const char*

  cout << x << endl;
  cout << y << endl;
  cout << name << endl;
  }
  ```
  ```

Storage Classes

```
{
  int mount;
  auto int month;
}
```

- The example above defines two variables with the same storage class, auto can only be used within functions, i.e., local variables.

- The register Storage Class
- The register storage class is used to define local variables that should be stored in a register instead of RAM.

```
{
  register int  miles;
}
```

- The register should only be used for variables that require quick access such as counters.

Storage Classes

- The static Storage Class
- The static storage class instructs the compiler to keep a local variable in existence during the life-time of the program instead of creating and destroying it each time it comes into and goes out of scope.

- Therefore, making local variables static allows them to maintain their values between function calls.

- The static modifier may also be applied to global variables. When this is done, it causes that variable's scope to be restricted to the file in which it is declared.

- In C++, when static is used on a class data member, it causes only one copy of that member to be shared by all objects of its class.

Storage Classes

```
#include <iostream>
// Function declaration
void func(void);
static int count = 10; /* Global variable */
main()
{
   while(count--)
   {
      func();
   }
   return 0;
}
// Function definition
void func( void )
{
   static int i = 5; // local static variable
   i++;
   std::cout << "i is " << i ;
   std::cout << " and count is " << count << std::endl;
}
```

Storage Classes

- The extern Storage Class
- The extern storage class is used to give a reference of a global variable that is visible to ALL the program files.

- The extern modifier is most commonly used when there are two or more files sharing the same global variables or functions as explained below.

- First File: main.cpp

```
int count ;
extern void write_extern();
main()
{
   count = 5;
   write_extern();
}
```

- Second File: support.cpp

```
extern int count;
void write_extern(void)
{
   std::cout << "Count is " << count << std::endl;
}
```

To Run the program

```
g++  support.cpp main.cpp –o mytestfile
```

To run the program

```
./mytestfile
```

Storage Classes

- The mutable Storage Class
- The mutable specifier applies only to class objects.
- It allows a member of an object to override constness.
- That is, a mutable member can be modified by a const member function.

Operators

- Arithmetic Operators
- Relational Operators
- Logical Operators
- Bitwise Operators
- Assignment Operators
- Misc Operators

Reference variable

```
int y = 10;
int& x = y;
```

- x does NOT create a new variable.
- x becomes another name for y.
- Both refer to the same memory location.
- Must be initialized at the time of declaration

```
int a = 5;
int b = 6;
int& x = a;
x = b; // This assigns value of b to a, NOT make x refer to b
```

```
#include <iostream>
using namespace std;
int main() {
    int y = 10;
    int& x = y;
    cout << "y = " << y << endl;     //10
    cout << "x = " << x << endl;     //10
    x = 20;
    cout << "After modifying x:" << endl;
    cout << "y = " << y << endl;     //20
    cout << "x = " << x << endl;     //20
    return 0;
}
```

Differences

Reference	Pointer
Must be initialized	Can be uninitialized
Cannot be NULL	Can be NULL
Cannot change reference	Can point to another variable
No dereferencing needed	Needs `*` to access value

- When Do We Use References?
- Function parameters (call by reference)
- Operator overloading Returning large objects efficiently

Arithmetic Operators

Operator	Description	Example
+	Adds two operands	A + B will give 30
-	Subtracts second operand from the first	A - B will give -10
*	Multiplies both operands	A * B will give 200
/	Divides numerator by de-numerator	B / A will give 2
%	Modulus Operator and remainder of after an integer division	B % A will give 0
++	Increment operator, increases integer value by one	A++ will give 11
--	Decrement operator, decreases integer value by one	A-- will give 9

Relational Operators

Operator	Description	Example
==	Checks if the values of two operands are equal or not, if yes then condition becomes true.	(A == B) is not true.
!=	Checks if the values of two operands are equal or not, if values are not equal then condition becomes true.	(A != B) is true.
>	Checks if the value of left operand is greater than the value of right operand, if yes then condition becomes true.	(A > B) is not true.
<	Checks if the value of left operand is less than the value of right operand, if yes then condition becomes true.	(A < B) is true.
>=	Checks if the value of left operand is greater than or equal to the value of right operand, if yes then condition becomes true.	(A >= B) is not true.
<=	Checks if the value of left operand is less than or equal to the value of right operand, if yes then condition becomes true.	(A <= B) is true.

Logical Operators

Operator	Description	Example
&&	Called Logical AND operator. If both the operands are non-zero, then condition becomes true.	(A && B) is false.
||	Called Logical OR Operator. If any of the two operands is non-zero, then condition becomes true.	(A || B) is true.
!	Called Logical NOT Operator. Use to reverses the logical state of its operand. If a condition is true, then Logical NOT operator will make false.	!(A && B) is true.

Bitwise Operators

Operator	Description	Example
&	Binary AND Operator copies a bit to the result if it exists in both operands.	(A & B) will give 12 which is 0000 1100
|	Binary OR Operator copies a bit if it exists in either operand.	(A | B) will give 61 which is 0011 1101
^	Binary XOR Operator copies the bit if it is set in one operand but not both.	(A ^ B) will give 49 which is 0011 0001
~	Binary Ones Complement Operator is unary and has the effect of 'flipping' bits.	(~A) will give -61 which is 1100 0011 in 2's complement form due to a signed binary number.
<<	Binary Left Shift Operator. The left operands value is moved left by the number of bits specified by the right operand.	A << 2 will give 240 which is 1111 0000
>>	Binary Right Shift Operator. The left operands value is moved right by the number of bits specified by the right operand.	A >> 2 will give 15 which is 0000 1111

Bitwise Operators

Bitwise operator works on bits and perform bit-by-bit operation. The truth tables for &, |, and ^ are as follows:

p	q	p & q	p | q	p ^ q
0	0	0	0	0
0	1	0	1	1
1	1	1	1	0
1	0	0	1	1

Assume if A = 60; and B = 13; now in binary format they will be as follows:

A = 0011 1100

B = 0000 1101

A&B = 0000 1100

A|B = 0011 1101

A^B = 0011 0001

~A = 1100 0011

Assignment Operators

Operator	Description	Example
=	Simple assignment operator, Assigns values from right side operands to left side operand	C = A + B will assign value of A + B into C
+=	Add AND assignment operator, It adds right operand to the left operand and assign the result to left operand	C += A is equivalent to C = C + A
-=	Subtract AND assignment operator, It subtracts right operand from the left operand and assign the result to left operand	C -= A is equivalent to C = C - A
*=	Multiply AND assignment operator, It multiplies right operand with the left operand and assign the result to left operand	C *= A is equivalent to C = C * A
/=	Divide AND assignment operator, It divides left operand with the right operand and assign the result to left operand	C /= A is equivalent to C = C / A
%=	Modulus AND assignment operator, It takes modulus using two operands and assign the result to left operand	C %= A is equivalent to C = C % A
<<=	Left shift AND assignment operator	C <<= 2 is same as C = C << 2
>>=	Right shift AND assignment operator	C >>= 2 is same as C = C >> 2
&=	Bitwise AND assignment operator	C &= 2 is same as C = C & 2
^=	bitwise exclusive OR and assignment operator	C ^= 2 is same as C = C ^ 2
|=	bitwise inclusive OR and assignment operator	C |= 2 is same as C = C | 2

Misc Operators

Operator	Description
sizeof	sizeof operator returns the size of a variable. For example, sizeof(a), where a is integer, will return 4.
Condition ? X : Y	Conditional operator. If Condition is true ? then it returns value X : otherwise value Y
,	Comma operator causes a sequence of operations to be performed. The value of the entire comma expression is the value of the last expression of the comma-separated list.
. (dot) and -> (arrow)	Member operators are used to reference individual members of classes, structures, and unions.
Cast	Casting operators convert one data type to another. For example, int(2.2000) would return 2.
&	Pointer operator & returns the address of an variable. For example &a; will give actual address of the variable.
*	Pointer operator * is pointer to a variable. For example *var; will pointer to a variable var.

Control Statements

Loops

- While
- Do..while
- For

Decision Making

- If
- If…else
- Nested If else
- Switch …case

- Break
- Continue

Functions

- Declaration
- Definition
- Calling function

```
#include<iostream>
using namespace std;
int square (int);
int main ()
 {
int z = 4;
cout << square(z);
}
int square (int x)
{   x = (x*x); return x;
}
```

Pass by value

```
void swap1(int x,int y)
{
    int temp=x;
    x = y;
    y=temp;
}
```

Pass by reference

```
void swap2(int& x,int& y)
{
    int temp=x;
    x = y;
    y=temp;
}
```

Arrays

- Definition
  - Int a[10]; //int[10] a;
  - Char b[12];

  - a[0]=10;
  - Cout<<a[0];
  - Int x[]={10,20,30};

Arrays

- int x[7];

x[0]	x[1]	x[2]	x[3]	x[4]	x[5]	x[6]

- int score[3][3]={{1,2,3},{2,3,4}{3,5,6}};

Pointer

- Example
  - int *p, char * s;
- The value of a pointer is just an address.
- Why pointers?
  - Dereferencing (*)
    - Get the content
  - Referencing (&)
    - Get the address of

Examples of pointer

- int *p;
- Int a;
- a=10;
- p=&a;
- *p=7;
- Int b=*p;

Dynamic allocating memory

- new , delete
- int *p=new int;
- int *p=new int [12];
- delete p;
- delete []p;
- malloc,…

Structure

```
struct person
{   long nId;
    char strName[30];
    int nAge;
    float fSalary;
    char strAddress[100];
    char strPhone[20];
};

struct person a ,  b, c;
struct person *p;
```

union

```
union num
{
   int x;
   float y;
}
```

More in a stucture: operations

```
struct box
{

    double dLength,dWidth,dHeight;
    double dVolume;

    double get_vol()
      {
        return dLength * dWidth * dHeight;
      }
}
```